Complete Guide to Tailwind CSS

WEB DESIGNING • CSS FRAMEWORK • BEGINNER TO ADVANCED

Tailwind CSS is a utility-first CSS framework used to design attractive, responsive and modern websites quickly.

Instead of writing separate CSS classes in a stylesheet, Tailwind provides ready-made utility classes that you apply directly to HTML elements.

Illustrative examples: modern dashboard UI, responsive landing page, cards, buttons and typography.

1. What is Tailwind CSS?

Normally, using HTML and CSS:

HTML

```
<h1 class="heading">Hello World</h1>
```

CSS

```
.heading {
    color: blue;
    font-size: 32px;
    font-weight: bold;
    text-align: center;
}
```

Using Tailwind CSS:

```
<h1 class="text-blue-600 text-3xl font-bold text-center">
    Hello World
</h1>
```

Here, you don't need to create a separate CSS class. Each Tailwind class represents a particular styling property.

2. Features of Tailwind CSS

• **Utility-first:** Provides small classes for individual CSS properties.

• **Responsive design:** Easily creates layouts for mobile, tablet and desktop.

• **Customization:** Allows custom colors, fonts, spacing and other styles.

• **Hover effects:** Provides classes for hover interactions.

• **Dark mode:** Supports dark and light themes.

• **Fast development:** Reduces the need to write custom CSS.

• **Reusable design:** Makes it easier to maintain consistent styling.

• **No predefined components:** Unlike Bootstrap, Tailwind primarily provides utility classes rather than ready-made UI components.

3. Tailwind CSS vs Bootstrap vs Normal CSS

Feature	Normal CSS	Bootstrap	Tailwind
Approach	Traditional CSS	Component-based	Utility-first
Ready-made components	No	Yes	No
Customization	High	High	High
Responsive design	Manual	Built-in	Built-in utilities
CSS writing	Separate file	Mostly classes	Classes in HTML

Feature	Normal CSS	Bootstrap	Tailwind
Learning	Easy	Easy	Moderate

Example:

• Bootstrap: `btn btn-primary`
• Tailwind: `bg-blue-600 text-white px-4 py-2 rounded`
• Normal CSS: Create your own class and CSS properties.

4. Installation of Tailwind CSS

Method 1: CDN (quick practice)

Add this inside your HTML <head>:

```
<script src="https://cdn.tailwindcss.com"></script>
```

Then write:

```
<h1 class="text-4xl text-blue-600 font-bold">
    Welcome to Tailwind CSS
</h1>
```

Note: This Play CDN approach is convenient for experimentation, but it is not recommended for production websites.

Method 2: Install using npm (Tailwind CSS v4)

```
npm install tailwindcss @tailwindcss/vite
```

Configure the Tailwind Vite plugin in your Vite configuration and import Tailwind in your CSS:

```
@import "tailwindcss";
```

Start your development server:

```
npm run dev
```

For a new Vite project, the recommended Tailwind v4 setup is:

1. Create a Vite project

```
npm create vite@latest my-project
cd my-project
npm install
```

2. Install Tailwind

```
npm install tailwindcss @tailwindcss/vite
```

3. Configure vite.config.js

```
import { defineConfig } from 'vite'
import tailwindcss from '@tailwindcss/vite'

export default defineConfig({
  plugins: [
    tailwindcss(),
  ],
})
```

4. In your CSS file

```
@import "tailwindcss";
```

5. Start the project

```
npm run dev
```

This is the current v4 Vite installation approach. For current details, see the official Tailwind documentation: https://tailwindcss.com/docs/installation/using-vite

Correction to the quick-practice CDN example above: for current Tailwind v4, use:

```
<script src="https://cdn.jsdelivr.net/npm/@tailwindcss/browser@4"></script>
```

5. Important Tailwind CSS classes

This is the most important section for practicals and examinations.

A. Text and typography

Class	Purpose
text-xs	Extra small text
text-sm	Small text
text-base	Normal text
text-lg	Large text
text-xl	Extra large text
text-2xl	2XL text
text-3xl	3XL text
text-4xl	4XL text
font-thin	Thin font
font-normal	Normal font
font-medium	Medium font
font-semibold	Semi-bold
font-bold	Bold
italic	Italic text
uppercase	Uppercase
lowercase	Lowercase
capitalize	Capitalize each word
underline	Underline
line-through	Strike-through
text-center	Center alignment
text-left	Left alignment
text-right	Right alignment

Example:

```
<h1 class="text-4xl font-bold text-center text-blue-600">
    Welcome to Tailwind
</h1>

<p class="text-lg italic text-gray-500">
    Learn Tailwind CSS
</p>
```

B. Background colors

Class	Purpose
bg-red-500	Red background
bg-blue-500	Blue background
bg-green-500	Green background
bg-yellow-500	Yellow background
bg-purple-500	Purple background

Class	Purpose
bg-pink-500	Pink background
bg-black	Black background
bg-white	White background
bg-gray-200	Light gray background
bg-gray-800	Dark gray background
bg-transparent	Transparent background

Example:

```
<div class="bg-blue-500">
    Blue Background
</div>

<div class="bg-red-100">
    Light Red Background
</div>
```

D. Margin and padding

Class	Meaning
m-4	Margin all sides
mt-4	Margin top
mb-4	Margin bottom
ml-4	Margin left
mr-4	Margin right
mx-4	Margin left and right
my-4	Margin top and bottom
p-4	Padding all sides
pt-4	Padding top
pb-4	Padding bottom
px-4	Padding left and right
py-4	Padding top and bottom

Example:

```
<div class="m-4 p-6 bg-blue-100">
    Hello Tailwind
</div>
```

Margin: Space outside an element. **Padding**: Space inside an element.

E. Width and height

Class	Purpose
w-full	100% width
w-1/2	50% width
w-1/3	One-third width
w-screen	Viewport width
w-64	Fixed width
h-full	100% height

Class	Purpose
h-screen	Viewport height
h-64	Fixed height
min-h-screen	Minimum screen height
max-w-xl	Maximum width XL

Example:

```
<div class="w-1/2 h-64 bg-green-500">
    Width 50%, Height 16rem
</div>
```

F. Border and border radius

Class	Purpose
border	Add border
border-2	2px border
border-4	4px border
border-red-500	Red border
rounded	Small rounded corners
rounded-md	Medium rounded corners
rounded-lg	Large rounded corners
rounded-xl	Extra large rounded corners
rounded-full	Fully rounded

Example:

```
<div class="border-2 border-blue-500 rounded-xl p-4">
    Tailwind Border Example
</div>
```

G. Display properties

Class	CSS property
block	display: block
inline	display: inline
inline-block	display: inline-block
flex	display: flex
grid	display: grid
hidden	display: none

Example:

```
<div class="flex">
    <div class="bg-red-500 p-4">One</div>
    <div class="bg-blue-500 p-4">Two</div>
</div>
```

H. Flexbox

Class	Purpose
flex	Enable flexbox
flex-row	Horizontal direction

Class	Purpose
flex-col	Vertical direction
justify-center	Center horizontally along main axis
justify-between	Space between items
justify-around	Space around items
justify-evenly	Equal spacing
items-center	Align items center
items-start	Align items at start
items-end	Align items at end
flex-wrap	Allow wrapping
gap-4	Gap between items

Example:

```
<div class="flex justify-center items-center gap-4">
    <div class="bg-red-500 p-4">Box 1</div>
    <div class="bg-blue-500 p-4">Box 2</div>
    <div class="bg-green-500 p-4">Box 3</div>
</div>
```

I. CSS Grid

Class	Purpose
grid	Enable grid
grid-cols-1	One column
grid-cols-2	Two columns
grid-cols-3	Three columns
grid-cols-4	Four columns
grid-rows-2	Two rows
col-span-2	Span two columns
gap-4	Grid gap

Example:

```
<div class="grid grid-cols-3 gap-4">
    <div class="bg-red-500 p-5">1</div>
    <div class="bg-blue-500 p-5">2</div>
    <div class="bg-green-500 p-5">3</div>
</div>
```

J. Position properties

Class	Purpose
static	Normal positioning
relative	Relative positioning
absolute	Absolute positioning
fixed	Fixed positioning
sticky	Sticky positioning
top-0	Top 0
bottom-0	Bottom 0

Class	Purpose
left-0	Left 0
right-0	Right 0
z-10	Z-index 10

K. Shadow and opacity

Class	Purpose
shadow-sm	Small shadow
shadow-md	Medium shadow
shadow-lg	Large shadow
shadow-xl	Extra large shadow
opacity-50	50% opacity
opacity-75	75% opacity
opacity-100	100% opacity

L. Hover effects

Prefix	Purpose
hover:	Mouse hover
focus:	Element receives focus
active:	Element is being pressed
disabled:	Disabled element
first:	First child
last:	Last child
odd:	Odd-numbered child
even:	Even-numbered child
group-hover:	Parent hover affects child

Example:

```
<button class="bg-blue-500 text-white px-6 py-3 rounded
               hover:bg-blue-700">
    Click Me
</button>
```

Tailwind uses the `hover:` prefix to apply styles when the user moves the mouse over an element. These variants can also be combined.

M. Responsive design

Prefix	Default screen width
sm:	640px
md:	768px
lg:	1024px
xl:	1280px
2xl:	1536px

Example:

```
<div class="bg-red-500
            sm:bg-blue-500
            md:bg-green-500
            lg:bg-yellow-500
            xl:bg-purple-500">

    Responsive Background

</div>
```

Here: Mobile: Red; Small screen and above: Blue; Medium screen and above: Green; Large screen and above: Yellow; Extra large screen and above: Purple.

Important: Tailwind follows a mobile-first approach. Unprefixed classes apply by default, and breakpoint-prefixed classes apply at that breakpoint and above.

N. Transition and animation

Class	Purpose
transition	Enable transitions
transition-all	Transition all properties
duration-300	300ms duration
ease-in	Ease-in effect
ease-out	Ease-out effect
animate-spin	Spinning animation
animate-ping	Ping animation
animate-pulse	Pulsing animation
animate-bounce	Bouncing animation

Example:

```
<button class="bg-purple-500 text-white p-4 rounded-lg
               transition duration-300 hover:bg-purple-800">
    Hover Me
</button>
```

C. Text colors

```
<p class="text-red-500">Red Text</p>
<p class="text-blue-500">Blue Text</p>
<p class="text-green-500">Green Text</p>
<p class="text-purple-500">Purple Text</p>
<p class="text-gray-500">Gray Text</p>
```

O. Dark mode

```
<div class="bg-white text-black dark:bg-gray-900 dark:text-white p-6">
    Dark Mode Example
</div>
```

Use `dark:` to apply styles in dark mode. Tailwind v4 supports dark-mode styling and configurable dark-mode behavior.

6. Useful Tailwind concepts

Concept	Explanation
Utility classes	Small classes for individual CSS properties
Variants	Prefixes like hover:, md:, dark:
Breakpoints	Screen sizes for responsive styling
Arbitrary values	Custom values using square brackets

Concept	Explanation
Theme variables	Define reusable design values
Preflight	Tailwind's base style reset
Content detection	Finds classes used in project files
Plugins	Extend Tailwind functionality

Arbitrary values example:

```
<div class="w-[350px] h-[200px] bg-[#ff5733]">
    Custom Size and Color
</div>
```

You can use custom values when the standard utility scale does not provide what you need.

7. Complete Tailwind CSS practical example

Let's create a responsive student profile card using Tailwind.

Illustrative Tailwind-style profile card.

Complete HTML code

```
<!DOCTYPE html>
<html>
<head>
    <title>Tailwind Profile Card</title>
    <script src="https://cdn.jsdelivr.net/npm/@tailwindcss/browser@4"></script>
</head>

<body class="bg-gray-100 flex justify-center items-center min-h-screen">

    <div class="bg-white shadow-xl rounded-2xl p-8
                w-80 text-center">

        <img src="https://placehold.co/100"
             class="rounded-full mx-auto mb-4">

        <h1 class="text-2xl font-bold text-blue-600">
            Student Profile
        </h1>

        <h2 class="text-lg font-semibold text-gray-700">
            Web Designer
        </h2>

        <p class="text-gray-500 mt-3">
            Learning HTML, CSS and Tailwind CSS.
        </p>

        <button class="bg-blue-600 text-white px-6 py-2
                       rounded-lg mt-5
                       hover:bg-blue-800
                       transition duration-300">
```

```
            View Profile
        </button>

    </div>

</body>
</html>
```

8. Tailwind CSS customization (v4)

Tailwind v4 allows you to customize theme values directly in CSS.

```
@import "tailwindcss";

@theme {
    --color-primary: #2563eb;
    --color-secondary: #9333ea;
    --font-display: Arial, sans-serif;
    --breakpoint-xs: 30rem;
}
```

You can then use the custom utilities:

```
<h1 class="text-primary font-display">
    Hello World
</h1>

<div class="bg-secondary">
    Custom Background
</div>
```

9. Advantages and disadvantages

Advantages

• Faster development.

• Less need for separate CSS.

• Easy responsive design.

• Highly customizable.

• Consistent spacing and colors.

• Good for modern web applications.

Disadvantages

• HTML can become lengthy due to many classes.

• Utility classes require practice to remember.

• Custom designs can initially take more time.

• Requires a build setup for production.

• Can be difficult to maintain if classes are not organized properly.

10. Tailwind CSS important practical topics

These are useful exercises to practise in your Web Designing Lab.

No.	Practical
1	Center a heading using Tailwind
2	Create a colored background
3	Apply different font sizes
4	Create a student profile card

No.	Practical
5	Create a pricing card
6	Design a navigation bar
7	Create a responsive grid
8	Create a responsive flex layout
9	Apply margin and padding
10	Create rounded buttons
11	Apply hover effects
12	Create a login form
13	Create a registration form
14	Create a responsive image gallery
15	Create a responsive website
16	Apply shadows and borders
17	Create a newsletter signup box
18	Create a product card
19	Apply dark mode
20	Create an animated button
21	Create a sidebar
22	Create a dashboard
23	Create a responsive footer
24	Create a landing page
25	Create a complete portfolio website

11. Important viva questions and answers

Q1. What is Tailwind CSS?
Tailwind CSS is a utility-first CSS framework used to design responsive websites using predefined utility classes.

Q2. What is utility-first CSS?
It is an approach in which small utility classes are used to style individual CSS properties directly in HTML.

Q3. Is Tailwind a CSS framework?
Yes, Tailwind is a utility-first CSS framework.

Q4. What is the difference between Bootstrap and Tailwind?
Bootstrap provides predefined UI components, whereas Tailwind provides utility classes to build custom designs.

Q5. What is the use of flex?
It enables the CSS Flexbox layout.

Q6. What is the use of grid?
It enables CSS Grid layout.

Q7. What is the use of p-4?
It applies padding on all four sides.

Q8. What is the use of m-4?
It applies margin on all four sides.

Q9. What is the use of hover:?
It applies styling when an element is hovered over.

Q10. What is the use of md:?
It applies styling at medium screen sizes and above.

Q11. What is the use of rounded-full?
It creates fully rounded corners, often used for circular images.

Q12. What is the use of shadow-lg?
It applies a large box shadow.

Q13. What is the use of hidden?
It hides an element using display: none.

Q14. What is the use of dark:?
It applies styles in dark mode.

Q15. Is JavaScript required in Tailwind CSS?
No. Tailwind is a CSS framework. JavaScript can be used separately for interactive functionality.

12. Important Tailwind reference resources

Official Tailwind CSS Documentation
https://tailwindcss.com/docs

Tailwind Play
https://play.tailwindcss.com/

Tailwind GitHub
https://github.com/tailwindlabs/tailwindcss

13. Recommended learning order

Basics
Text, colors, fonts, backgrounds and borders.

Spacing and sizing
Margin, padding, width and height.

Layout
Display, Flexbox, Grid and positioning.

Design effects
Shadows, rounded corners, transitions and animations.

Advanced styling
Hover, focus, responsive design and dark mode.

Projects
Cards, forms, navbar, dashboard and portfolio.

My recommendation: Practise all 25 exercises in section 10. They will help you understand almost all the commonly used Tailwind utilities.